JavaScript Object Practice

20 calculation-based exercises • Object keys, data types, and formulas

Instructions: For each task, create a JavaScript object with the listed properties. Use the formula to calculate the required result, then display the object details and calculated value using `console.log()`.

1. Product Discount Calculator

Task: Calculate the final discounted price of a product with inventory details.

Object keys and data types: productId (Number); title (String); category (String); price (Number); discountPercent (Number); stockCount (Number); rating (Number)

Formula:

```
Final price = price - (price * discountPercent / 100)
```

2. Employee Salary with Bonus

Task: Calculate total salary including bonus rate and allowances.

Object keys and data types: empId (Number); name (String); department (String); designation (String); baseSalary (Number); bonusRate (Number); allowance (Number)

Formula:

```
Total salary = baseSalary + (baseSalary * bonusRate) + allowance
```

3. Car Rental Total Rent with Tax

Task: Calculate total car rental cost including insurance and tax.

Object keys and data types: carId (String); model (String); rentPerDay (Number); days (Number); insuranceFee (Number); taxPercent (Number)

Formula:

```
Total = (rentPerDay * days) + insuranceFee + (((rentPerDay * days) + insuranceFee) *
taxPercent / 100)
```

4. Rectangle Area & Perimeter

Task: Calculate area and perimeter using geometric properties.

Object keys and data types: shapeId (String); shapeName (String); length (Number); width (Number); unit (String); color (String)

Formula:

```
Area = length * width
```

```
Perimeter = 2 * (length + width)
```

5. Simple Interest Calculator

Task: Calculate simple interest with loan metadata and a processing fee.

Object keys and data types: loanId (String); borrowerName (String); principal (Number); rate (Number); years (Number); processingFee (Number)

Formula:

```
Interest + fee = ((principal * rate * years) / 100) + processingFee
```

6. Shopping Cart Total with GST

Task: Calculate cart subtotal with shipping and GST.

Object keys and data types: cartId (Number); itemName (String); itemPrice (Number); quantity (Number); shippingFee (Number); gstRate (Number)

Formula:

```
Total = (itemPrice * quantity) + shippingFee + (((itemPrice * quantity) + shippingFee) *
gstRate / 100)
```

7. Temperature Conversion (Celsius to Fahrenheit)

Task: Convert temperature with related weather data.

Object keys and data types: stationId (Number); cityName (String); celsius (Number); humidity (Number); windSpeed (Number)

Formula:

```
Fahrenheit = (celsius * 9 / 5) + 32
```

8. Net Salary After Deductions

Task: Calculate net salary after tax, PF, and insurance deductions.

Object keys and data types: empId (String); name (String); grossSalary (Number); taxRate (Number); providentFund (Number); insuranceDeduction (Number)

Formula:

```
Net salary = grossSalary - (grossSalary * taxRate / 100) - providentFund - insuranceDeduction
```

9. Fuel Cost Calculator

Task: Calculate trip fuel cost including toll charges.

Object keys and data types: vehicleNo (String); driverName (String); totalKm (Number); mileage (Number); pricePerLiter (Number); tollCharges (Number)

Formula:

```
Total = (totalKm / mileage * pricePerLiter) + tollCharges
```

10. Body Mass Index (BMI) Calculator

Task: Calculate BMI using a person's record details.

Object keys and data types: patientId (Number); name (String); age (Number); weightKg (Number); heightM (Number); gender (String)

Formula:

```
BMI = weightKg / (heightM * heightM)
```

11. Restaurant Bill with Tip and Tax

Task: Calculate the restaurant bill with tip and tax amounts.

Object keys and data types: billNo (Number); waiterName (String); foodAmount (Number); tipPercent (Number); taxPercent (Number)

Formula:

```
Total = foodAmount + (foodAmount * tipPercent / 100) + (foodAmount * taxPercent / 100)
```

12. Electricity Bill Calculation

Task: Calculate the electricity bill using units, rate, fixed charge, and meter rent.

Object keys and data types: consumerNo (String); consumerName (String); units (Number); ratePerUnit (Number); fixedCharge (Number); meterRent (Number)

Formula:

```
Total = (units * ratePerUnit) + fixedCharge + meterRent
```

13. Bank Account Balance Update

Task: Calculate the updated bank balance after transactions and a maintenance fee.

Object keys and data types: accountNo (String); accountHolder (String); balance (Number); deposit (Number); withdrawal (Number); maintenanceFee (Number)

Formula:

```
Updated balance = balance + deposit - withdrawal - maintenanceFee
```

14. Student Average Marks Calculator

Task: Calculate average marks from a subject marks array and practical marks.

Object keys and data types: rollNo (Number); name (String); std (Number); marks (Array of Numbers); practicalMarks (Number)

Formula:

```
Given formula = ((marks[0] + marks[1] + marks[2] + marks[3] + marks[4]) / marks.length) +
practicalMarks
```

15. Cuboid Volume Calculator

Task: Calculate cuboid volume using its dimensions and material details.

Object keys and data types: boxId (String); material (String); length (Number); width (Number); height (Number); density (Number)

Formula:

```
Volume = length * width * height
```

16. Currency Converter (USD to INR)

Task: Convert USD to INR including a transaction fee.

Object keys and data types: transactionId (String); currencyName (String); usdAmount (Number); exchangeRate (Number); processingFee (Number)

Formula:

```
Total in INR = (usdAmount * exchangeRate) + processingFee
```

17. Project Total Cost with Maintenance

Task: Calculate the project budget including design and maintenance charges.

Object keys and data types: projectId (String); clientName (String); developmentCost (Number); designCost (Number); maintenancePercent (Number)

Formula:

```
Total = (developmentCost + designCost) + ((developmentCost + designCost) * maintenancePercent
/ 100)
```

18. Asset Depreciation Value

Task: Calculate current machinery value after depreciation.

Object keys and data types: assetId (String); assetName (String); originalPrice (Number); depreciationRatePerYear (Number); yearsUsed (Number); salvageValue (Number)

Formula:

```
Given formula = originalPrice - ((originalPrice * depreciationRatePerYear * yearsUsed) / 100)
+ salvageValue
```

19. Circle Area & Circumference

Task: Calculate circle area and circumference using its radius.

Object keys and data types: circleId (String); color (String); radius (Number); pi (Number); thickness (Number)

Formula:

```
Area = pi * radius * radius
```

```
Circumference = 2 * pi * radius
```

20. Business Net Profit Calculator

Task: Calculate net profit after expenses, taxes, and marketing costs.

Object keys and data types: companyName (String); financialYear (String); totalRevenue (Number); totalExpenses (Number); taxAmount (Number); marketingCost (Number)

Formula:

```
Net profit = totalRevenue - totalExpenses - taxAmount - marketingCost
```